

CURE-BENCH CHALLENGE

MEDICAL REASONING



Team MedAI: DailyMed Integration for TxAgent

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CURE-BENCH CHALLENGE

Therapeutic decision-making in clinical medicine constitutes a high-stakes domain in which AI guidance interacts with complex interactions among patient characteristics, disease processes, and pharmacological agents. Tasks such as drug recommendation, treatment planning, and adverse-effect prediction demand robust, multi-step reasoning grounded in reliable biomedical knowledge. Agentic AI methods, exemplified by TxAgent, address these challenges through iterative retrieval-augmented generation (RAG). TxAgent employs a finetuned Llama-3.1-8B model that dynamically generates and executes function calls to a unified biomedical tool suite (ToolUniverse), integrating FDA Drug API, OpenTargets, and Monarch resources to ensure access to current therapeutic information. In contrast to general-purpose RAG systems, medical applications impose stringent safety constraints, rendering the accuracy of both the reasoning trace and the sequence of tool invocations critical.

APPROACH

We chose TxAgent as our baseline Agentic Agent and integrated DailyMed as an additional source of medical knowledge. Further, we compare multiple retrieval strategies, from BM25 to SotA dense retriever, outline LLM performance on fixed retrieval settings, and the influence of answer permutation on acc [%]. Additionally, we present our submission settings on all competition datasets, incorporating token-usage [n] and tool-call [n] metrics.

ABBREVIATIONS EXPLAINED

Based on the competition structure, we consider MC (multiple-choice) and OE-MC (open-ended multiple-choice) for each question in the validation dataset, using separate datasets with all 459 questions and answer ground truth labels.

Phase 1 (2097) and Phase 2 (2491) (competition evaluation) – set without ground-truth labels and separation.

RETRIEVAL A. GENERATION

TxAgent's retrieval system, finetuned Gwen2-1.5B, returns a set of 10 tool-call function names based on a generated retrieval query, considering the complete iterative built context, and achieves high performance (Fig. 1). Non-finetuned SotA dense retriever stabilizes in performance on OE-MC questions; MC question accuracy decreases continuously. Sparse and no retrieval are outperformed.

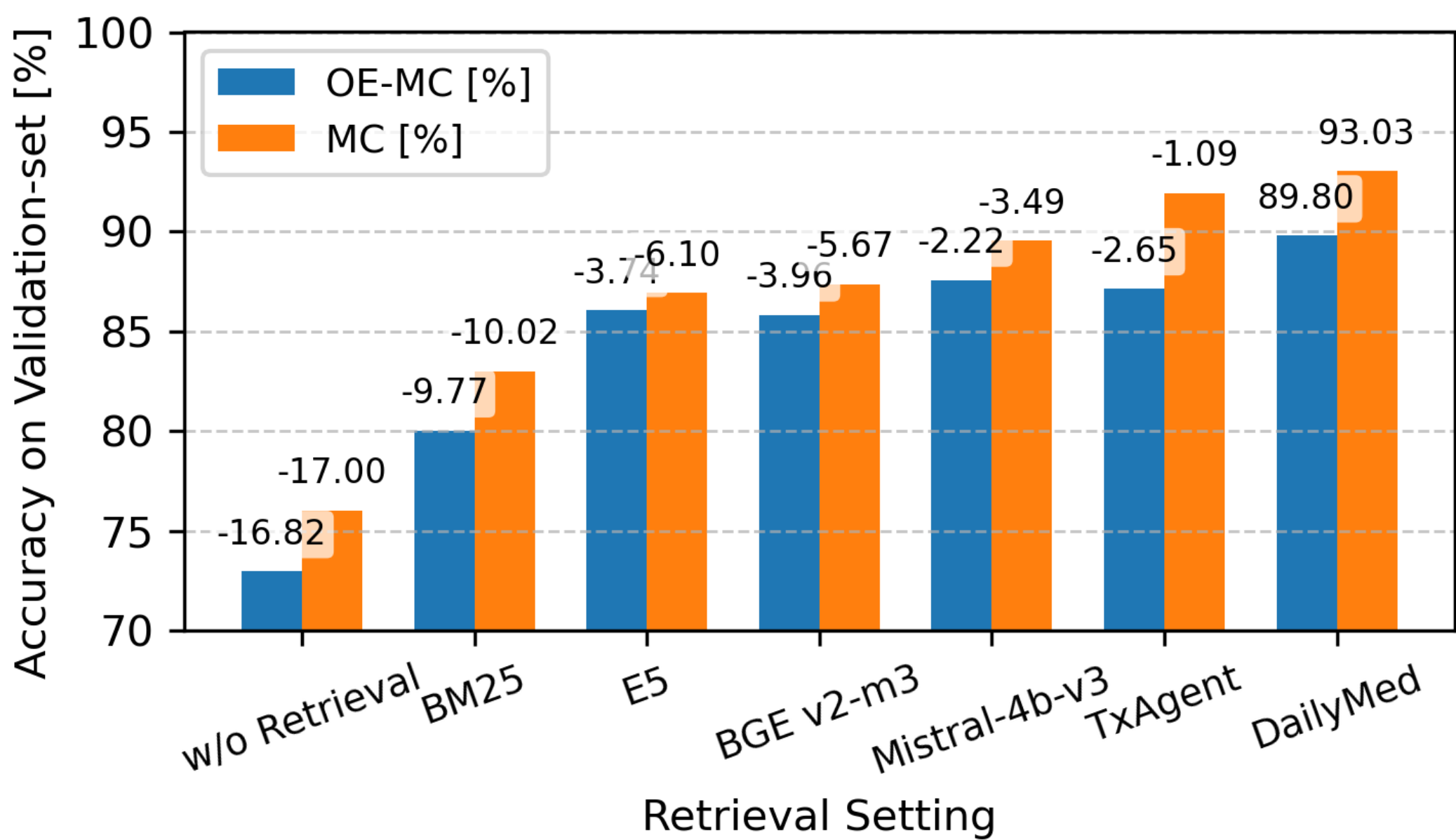


Fig 1: Retriever comparison using k@10 for all settings.

Setting	Avg.			
	Tokens in [10 ³]	Tokens out [n]	Func. calls [n]	Acc [%]
<i>w/ Summary Agent</i>				
<i>Validation-set</i>				
TxAgent (OE-MC)	32.3	780	3.33	88.02
TxAgent, w/ DailyMed (OE-MC)	24.6	601	3.60	89.80
TxAgent (MC)	23.1	571	2.89	91.29
TxAgent, w/ DailyMed (MC)	17.1	462	3.21	93.03
<i>Phase 1-set</i>				
TxAgent, Max Rounds 15, w/o DailyMed	60.0	1074	5.54	58.8
<i>Phase 2-set</i>				
TxAgent, Max Rounds 10, w/ DailyMed	31.3 (In+Out)		3.55	64.67
TxAgent, w/ DailyMed	59.3 (In+Out)		3.84	64.80
TxAgent, w/ CallAgent, w/ DailyMed	205.5 (In+Out)		24.04	66.26

Fig 2: Performance and metric overview for submissions.

For all retrieval settings, the generated retrieval query compares to the function descriptions provided by ToolUniverse.

Not considering precision@k and recall@k, due to the absence of ground-truth tool-call labels per question.

VANILLA LLM PERFORMANCE

To investigate the performance of TxAgent's finetuned Llama3.1-8B, we define a single validation-set run with the corresponding retrieved context as a fixed ground-truth retrieval, described as "Tx + vanilla LLM". The LLM solves the RAG tasks with a simple prompt structure containing the retrieved context followed by the question. The complete overview is visualized in Fig.3 and Fig.4, combining the acc [%] for various open-source

LLMs for the described RAG setting and w/o retrieval, along with answer-option permutation, compared to the initial reference TxAgent acc [%].

DAILYMED INTEGRATION

A DailyMed JSON-formatted function description extends ToolUniverse's existing function descriptions. TxAgent's ToolRAG tool-retrieval procedure selects DailyMed without additional finetuning.

CONSIDERATION

TxAgent's retrieval system approaches tool-calling with precise tool-selection to minimize context extension, resulting in iterative function call executions. Our extension of DailyMed does not consider such a fine-grained selection. Therefore, retrieving information that would need several ToolUniverse function calls.

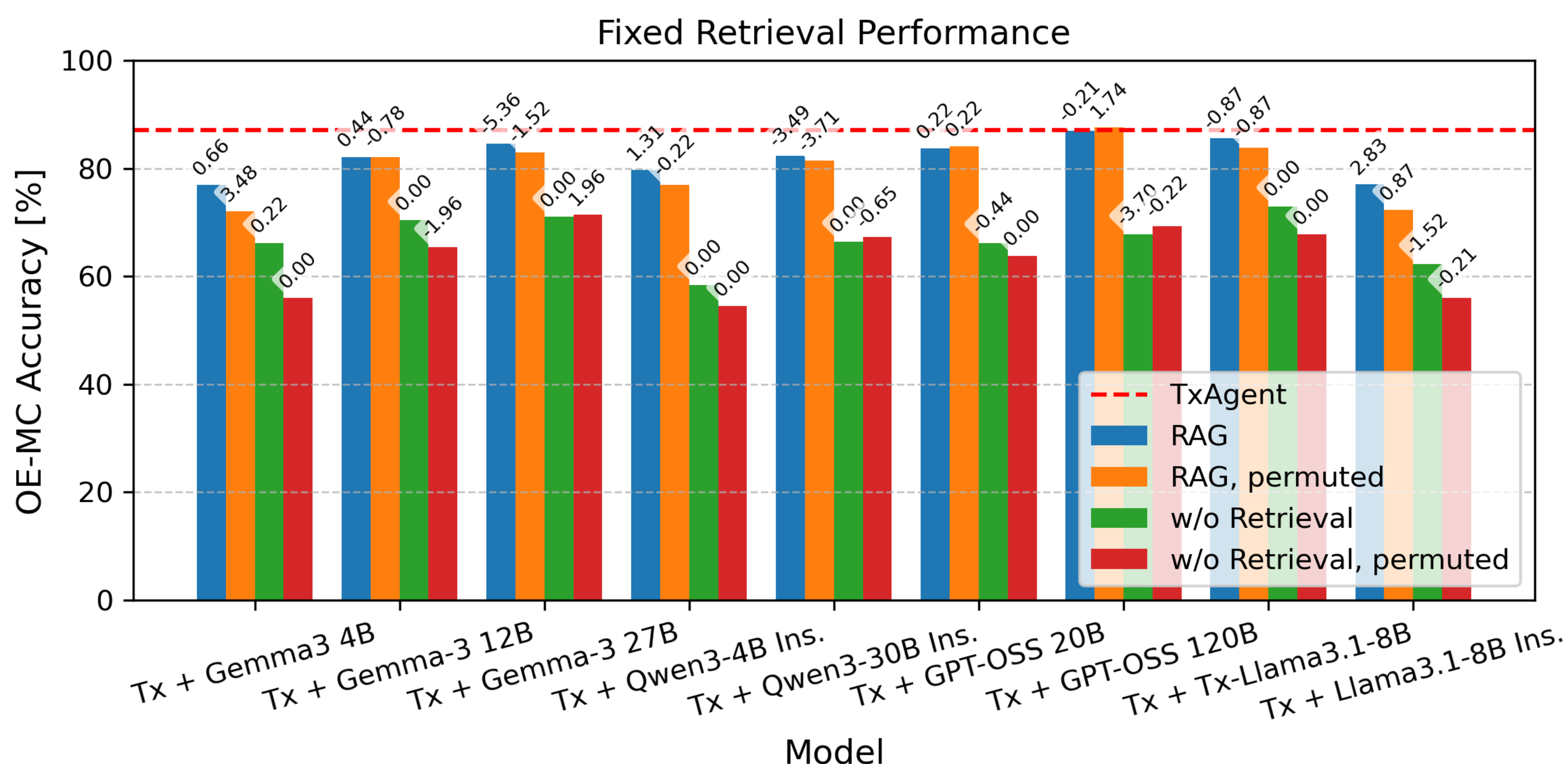
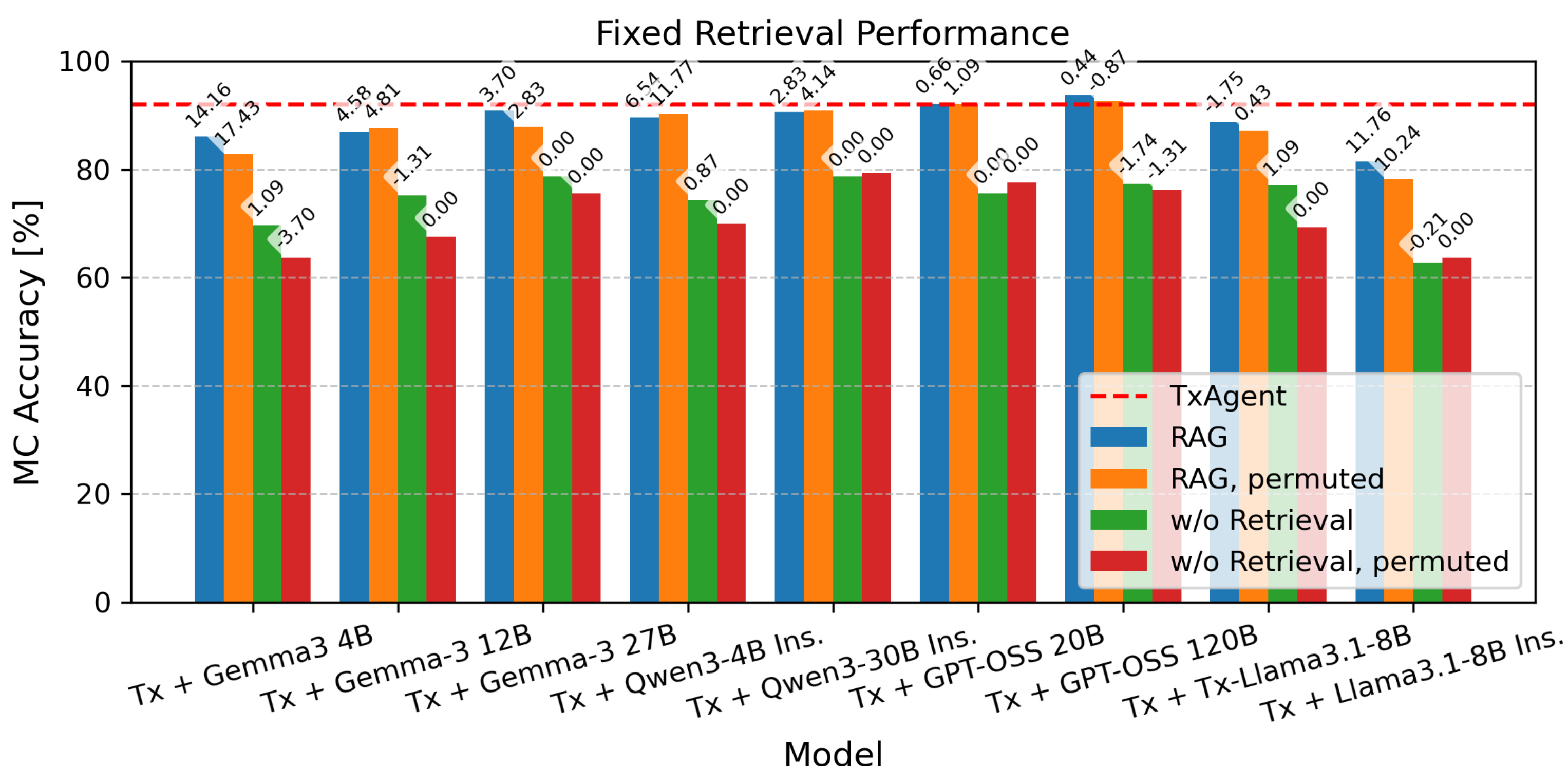


Fig 4,5: Fixed-retrieval performance comparison of TxAgent's retrieval with various LLMs and answer permutation on MC and OE-MC validation set.



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